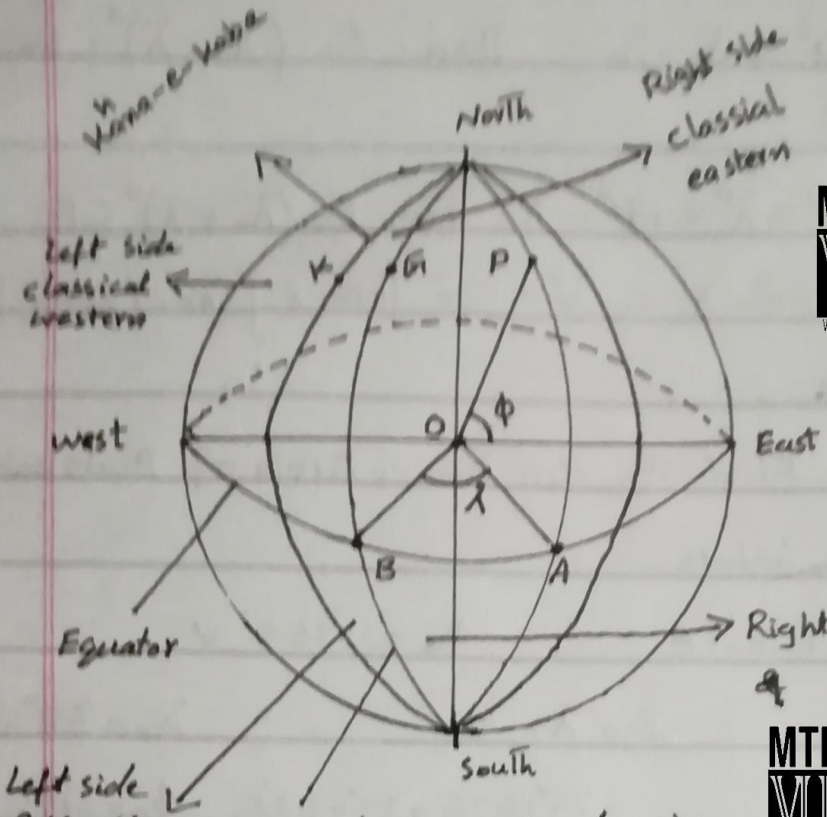
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Short Lecture
43,44 Longitude & Latitude QIBLA DIRECTION
SIR NAIMAT ULLAH



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Longitude = $AOB = \lambda$ $\quad P(\lambda, \phi)$

Latitude = $POE = \phi$

Longitude of Khana-e-Kaba = $\lambda_0 = 39^\circ 49' .2'' E$

Latitude of Khana-e-Kaba = $\phi_0 = 21^\circ 25' .2'' N$

Qibla direction

$$\tan i = P - Q \quad , \quad i = \tan^{-1}(P - Q)$$

$$P = \frac{\sin\phi}{\tan l} \quad , \quad Q = \frac{\cos\phi \tan\phi_0}{\sin l}$$

the classical longitude (l) of a place can be found by

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$\lambda^\circ E$ If $\lambda_0 < \lambda^\circ \leq 180^\circ$ Then $l = (\lambda - \lambda_0)^\circ CE$

If $0^\circ \leq \lambda^\circ < \lambda_0$ Then $l = (\lambda_0 - \lambda)^\circ CW$

$\lambda^\circ W$

If $0^\circ < \lambda^\circ < 180^\circ - \lambda_0$ Then $l = (\lambda_0 + \lambda)^\circ CE$

If $180^\circ - \lambda_0 < \lambda^\circ < 180^\circ$ Then $l = [360 - (\lambda_0 + \lambda)] CE$

Example

Find the direction of Qibla of Badshahi Mosque, Lahore

$\lambda_0 = 74^\circ 18.7' E$ $\phi_0 = 31^\circ 35.4' N$

Solution $l = \lambda - \lambda_0$ $\lambda_0 = 39^\circ 49.2' E$

$= 74^\circ 18.7' E - 39^\circ 49.2' E$ $\phi_0 = 21^\circ 25.2' N$

$= 34^\circ 29.5' CE$

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$$P = \frac{\sin\phi}{\tan l} = \frac{\sin(31^\circ 35.4')}{\tan(34^\circ 29.5')} = 0.7624$$

$$Q = \frac{\cos\phi \tan\phi_0}{\sin l} = \frac{\cos 31^\circ 35.4' \tan 21^\circ 25.2'}{\sin 34^\circ 29.5'} = [illegible]$$

$Q = 0.5901$

$i = \tan^{-1}(P - Q) = \tan^{-1}(0.7624 - 0.5901)$

$i = \tan^{-1}(0.1723)$

$i = 9^\circ 46.7'$ south of west

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